

# A Comparison of Statistical, Temporal, Spatial, and Spatio-Temporal Models for Electric Vehicle Charging Demand Forecasting

Thao Nguyen

thaonguyen.525202281449@st.ueh.edu.vn

## Abstract

The rapid growth of electric vehicles has increased the need for accurate charging demand forecasting at the urban charging-network level. Reliable forecasting can support charging infrastructure planning, demand management, and charging station operation. This study conducts a comparative evaluation of four forecasting approaches for electric vehicle charging occupancy prediction: ARIMA, LSTM, GCN, and GCN-LSTM. Using the UrbanEV dataset, the study finds that under the current setting, station-specific temporal patterns provide stronger predictive signals than the spatial dependencies. This suggests that more complex spatio-temporal models do not always guarantee better forecasting accuracy.

**Keywords:** electric vehicle charging demand forecasting; time-series forecasting; graph convolutional network; long short-term memory; spatio-temporal modeling; charging station occupancy.

# Contents

|          |                                             |          |
|----------|---------------------------------------------|----------|
| <b>1</b> | <b>Introduction</b>                         | <b>3</b> |
| <b>2</b> | <b>Comparative Methodologies</b>            | <b>3</b> |
| 2.1      | ARIMA . . . . .                             | 3        |
| 2.2      | LSTM - Long Short Term Memory . . . . .     | 4        |
| 2.3      | GCN - Graph Convolutional Network . . . . . | 4        |
| 2.4      | GCN-LSTM . . . . .                          | 5        |
| <b>3</b> | <b>Dataset</b>                              | <b>5</b> |
| <b>4</b> | <b>Results</b>                              | <b>6</b> |
| <b>5</b> | <b>Discussion</b>                           | <b>7</b> |

# 1 Introduction

Electric vehicles (EVs) have seen a significant increase in popularity recently due to the Sustainable Development Goals of the United Nations concerning carbon neutrality. Electric vehicle charging demand prediction is important for vacant charging pile recommendation and charging infrastructure planning (Kuang et al., 2025; Hüttel et al., 2021; Li et al., 2025).

The development of EV charging demand forecasting models has evolved from traditional statistical methods to advanced machine learning and deep learning approaches. Early studies employed statistical techniques such as Linear Regression (LR), Multiple Linear Regression (MLR), Spatial Autoregressive Regression (SAR), ARIMA, and SARIMA to model temporal demand patterns and spatial dependencies (ElGhanam et al., 2024; Koohfar et al., 2023). Subsequently, machine learning algorithms including Random Forest, XGBoost, and LightGBM demonstrated improved capability in capturing non-linear relationships between charging demand and contextual variables (Wang et al., 2025). Deep learning models, such as LSTM, further enhanced forecasting performance by learning complex temporal dependencies from historical charging data (Sanami et al., 2025). More recently, graph-based approaches have emerged to address the spatial nature of charging networks. For example, Tupayachi et al. (2025) proposed a Traffic-Weather Graph Convolutional Network (TW-GCN) that integrates charging, traffic, weather, and POI data to model spatio-temporal dependencies among charging stations. The success of these graph-based methods suggests the potential of jointly learning spatial and temporal relationships, making Graph Convolutional Networks (GCNs) and their variants promising solutions for EV charging demand forecasting.

In practical urban charging management, forecasting demand for a single charging station is often insufficient. Charging operators and city planners require a model that can provide accurate demand forecasts at the system level, covering a large network of charging stations across an entire city. Therefore, this study focuses not only on station-level prediction, but also on evaluating forecasting performance globally across the full set of charging stations in a city.

The following four distinct approaches for forecasting electric vehicle (EV) charging demand are considered in this study: statistical learning (ARIMA), temporal deep learning (LSTM), spatial deep learning (GCN), and spatio-temporal deep learning (GCN-LSTM).

## 2 Comparative Methodologies

### 2.1 ARIMA

For a specific charging station  $i \in V$ , the historical occupancy sequence extracted from the UrbanEV dataset is denoted as  $X^i = \{x_1^i, x_2^i, \dots, x_t^i\}$ , where  $x_t^i$  represents the occupancy level of station  $i$  at time step  $t$ .

Since ARIMA is a univariate time-series model, each charging station is modeled independently using only its optimal  $(p, d, q)$  using the `auto.arima` function.

## 2.2 LSTM - Long Short Term Memory

Long Short-Term Memory (LSTM) is used as the temporal deep learning model in this study. Unlike ARIMA, which models each charging station using a linear statistical formulation, LSTM is able to learn nonlinear temporal dependencies from historical occupancy sequences.

For a specific charging station  $i \in V$ , the historical occupancy sequence is denoted as  $X^i = \{x_1^i, x_2^i, \dots, x_t^i\}$ , where  $x_t^i$  represents the occupancy level of station  $i$  at time step  $t$ .

Given an input window of the previous  $T$  time steps, the LSTM model learns a mapping from the hour historical sequence  $\{x_{t-T+1}^i, x_{t-T+2}^i, \dots, x_t^i\}$  to the next-step occupancy value  $\hat{x}_{t+1}^i$ .

The LSTM model uses historical occupancy as the only input feature. The input sequence length is set to 12, corresponding to the previous 12 hourly observations. The model uses two LSTM layers with a hidden dimension of 64. The output representations across the 12 time steps are passed through the LSTM to predict the next 1 hour occupancy.

## 2.3 GCN - Graph Convolutional Network

The EV charging demand forecasting problem is presented via the occupancy (number of occupied charging slots) and formulated as a spatio-temporal graph learning task on a graph

$$G = (V, E, A)$$

where:

- $V = \{v_1, v_2, \dots, v_N\}$  denotes the set of EV charging stations.
- $E$  denotes the connectivity relationships between stations.
- $A \in \mathbb{R}^{N \times N}$  is the adjacency matrix describing spatial dependencies among stations.

Each node corresponds to one EV charging station and is associated with a historical occupancy sequence.

At time step  $t$ , the graph signal is represented as

$$X_t = [x_t^1, x_t^2, \dots, x_t^N]^T$$

where  $x_t^i$  denotes the occupancy level of station  $i$ .

Given the historical observations from the previous  $T$  time steps,

$$X_{t-T+1}, X_{t-T+2}, \dots, X_t,$$

the objective is to predict the occupancy at the next time step:

$$\hat{X}_{t+1} = f(X_{t-T+1:t}, A)$$

where  $f(\cdot)$  denotes the proposed GCN model.

The EV charging network is represented as a graph with 275 nodes, corresponding to 275 charging stations in Shenzhen, while the normalized adjacency matrix describes the spatial relationships among stations. For each node, occupancy values from the previous 12 time steps are used as input. The output sequence is 1, which means that occupancy data in the last 12 hours is utilized to forecast the occupancy of the next 1 hour. The model includes 2 layers of hidden dimension of 32.

## 2.4 GCN-LSTM

The GCN-LSTM model is used to capture both spatial and temporal dependencies in EV charging demand forecasting. While GCN learns spatial relationships among charging stations through the adjacency matrix, LSTM captures temporal patterns from historical occupancy sequences.

In this study, the GCN-LSTM model uses historical occupancy as the only input feature. The input sequence length is set to 12, corresponding to the previous 12 hourly observations. The GCN component consists of 2 graph convolution layers with a hidden dimension of 32 to learn spatial dependencies among 275 charging stations. The extracted spatial features are then fed into an LSTM layer to capture temporal patterns, followed by a fully connected layer to predict the next 1 hour occupancy.

## 3 Dataset

This research utilizes the dataset UrbanEV, an open dataset showcasing EV charging space availability and electricity consumption in a pioneering city for vehicle electrification, namely Shenzhen, China. UrbanEV offers a rich repository of charging data (i.e., charging occupancy, duration, volume, and price) captured at hourly intervals across an extensive six-month span for over 275 individual charging stations.

Table 1: Data Statistics

| Features                             | Max | Min | Median | Mean  |
|--------------------------------------|-----|-----|--------|-------|
| Occupancy (number of occupied slots) | 373 | 0   | 11     | 17.87 |
| Number of neighboring stations       | 11  | 1   | 5      | 5.35  |

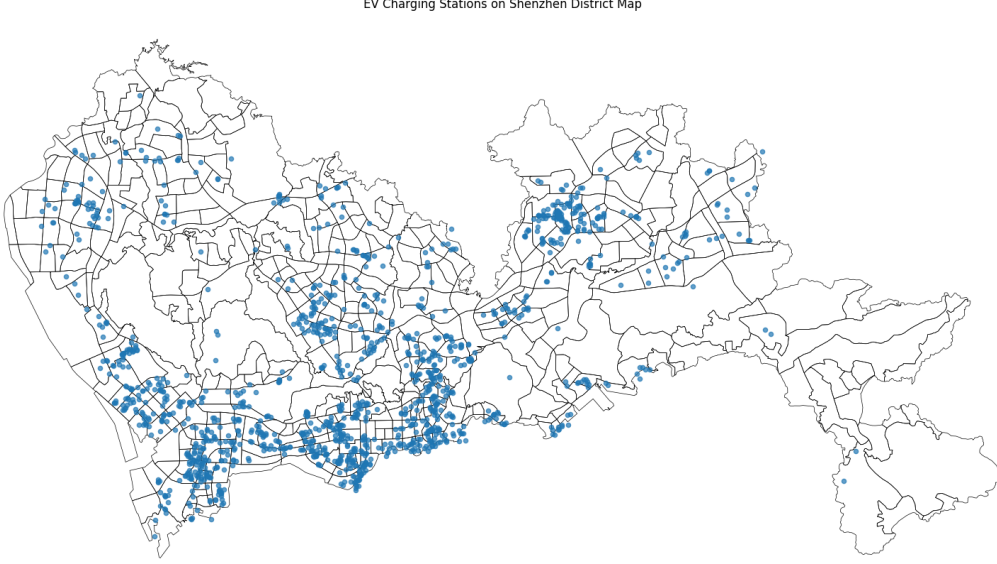


Figure 1: Spatial distribution of EV charging stations in Shenzhen District, China

## 4 Results

The results of all above algorithms are presented in the table below:

Table 2: Performance comparison of forecasting models

| Model    | MAE           | MSE            | RMSE          | Rank     |
|----------|---------------|----------------|---------------|----------|
| LSTM     | <b>1.8147</b> | <b>13.4340</b> | <b>3.6652</b> | <b>1</b> |
| GCN-LSTM | 4.8809        | 53.4173        | 7.3087        | 2        |
| GCN      | 5.3879        | 65.6817        | 8.1044        | 3        |
| ARIMA    | 6.5649        | 128.1186       | 11.3189       | 4        |

The experimental results show that LSTM achieves the best forecasting performance among all compared models, with the lowest MAE, MSE, and RMSE values. GCN-LSTM ranks second, outperforming the standalone GCN model, which indicates that combining temporal learning with spatial graph information can improve performance compared with spatial learning alone. However, the pure LSTM model still performs substantially better than the spatio-temporal GCN-LSTM model. This suggests that,

under the current experimental setting, temporal dependency is the dominant predictive signal for EV charging occupancy forecasting.

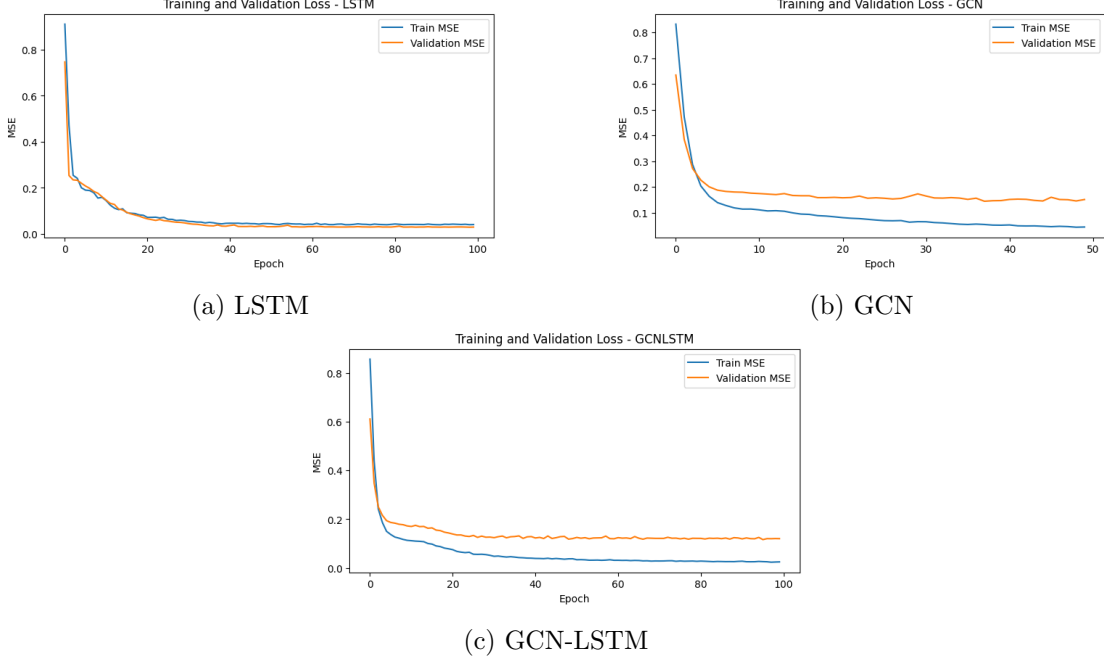


Figure 2: Training and validation loss curves of LSTM, GCN, and GCN-LSTM models

As shown in Figure 2, all three models demonstrate stable training and validation with no sight of overfitting due to both training and validation loss gradually decreasing over the epochs. However, LSTM is still the model with the highest similarity and narrowest gap between the training and validation loss, indicating better performance compared with other two deep learning forecasting models.

## 5 Discussion

Although spatio-temporal models are expected to perform better in this EVCS networked forecasting problems, the results of this study show that more complex model does not lead to higher accuracy. When demand is highly expressed by temporal patterns, temporal-only models may already capture most of the useful predictive information. Rodrigues (2022), for example, found that a simple model based on average weekly patterns and linear regression achieved performance comparable to several deep spatio-temporal models and even outperformed them on some datasets. This suggests that temporal patterns may be more important than spatial correlation in certain forecasting contexts.

The effectiveness of spatio-temporal forecasting also depends on the relationships among nodes. Nearby or connected locations often directly influence one another. Therefore, spatio-temporal models without strong spatial dependency that ignore it may lose important predictive signals (Jeon et al., 2025; Wang et al., 2023). However, EV charging demand at one station may not be strongly affected by occupancy at neighboring stations. Demand may instead be driven mainly by station-specific usage patterns.

Another possible explanation is that spatial dependence is not necessarily constant over time, but it may vary with travel flows, congestion, charging availability, and user substitution between stations (Jeon et al., 2025). However, the dataset uses a fixed adjacency matrix, which assumes that the relationships between charging stations remain unchanged all the time. Therefore, future research could consider adaptive adjacency matrices and add contextual variables such as traffic conditions, weather, points of interest, and charging prices.



## References

1. Cavus, M., Ayan, H., Dissanayake, D., Sharma, A., Deb, S., & Bell, M. (2025). Forecasting Electric Vehicle Charging Demand in Smart Cities Using Hybrid Deep Learning of Regional Spatial Behaviours. *Energies*, 18(13), 3425. <https://doi.org/10.3390/en18133425>
2. Elghanam, E., Alzaatreh, A., Hassan, M. S., & Osman, A. H. (2024). A Data-Driven Approach for EV Electricity Demand Modeling Using Spatial Regression: A UAE Case Study. *IEEE Access*, 12, 57302–57314. <https://doi.org/10.1109/ACCESS.2024.3390992>
3. Jeon, H., Lee, H., Kim, J., & Ko, S. (2025). ST-LINK: Spatially-aware large language models for spatio-temporal forecasting. In Proceedings of the 34th ACM International Conference on Information and Knowledge Management (pp. 10571067). Association for Computing Machinery. <https://doi.org/10.1145/3746252.3761085>
4. Koohfar, S., Woldemariam, W., & Kumar, A. (2023). Prediction of Electric Vehicles Charging Demand: A Transformer-Based Deep Learning Approach. *Sustainability*, 15(3), 2105. <https://doi.org/10.3390/su15032105>
5. Ostermann, A., & Haug, T. (2024). Probabilistic forecast of electric vehicle charging demand: Analysis of different aggregation levels and energy procurement. *Energy Informatics*, 7(1), 13. <https://doi.org/10.1186/s42162-024-00319-1>
6. Rodrigues, F. (2022). On the importance of stationarity, strong baselines and benchmarks in transport prediction problems [Preprint]. arXiv. <https://doi.org/10.48550/arXiv.2203.02954>
7. Sanami, S., Mosalli, H., Yang, Y., Yeh, H.-G., & Aghdam, A. G. (2025). Demand Forecasting for Electric Vehicle Charging Stations using Multivariate Time-Series Analysis (arXiv:2502.16365). *arXiv*. <https://doi.org/10.48550/arXiv.2502.16365>
8. Tran, M., Banister, D., Bishop, J. D. K., & McCulloch, M. D. (2012). Realizing the electric-vehicle revolution. *Nature Climate Change*, 2(5), 328–333. <https://doi.org/10.1038/nclimate1429>
9. Tupayachi, J., Camur, M. C., Heaslip, K., & Li, X. (2025). Spatio-Temporal Graph Convolutional Networks for EV Charging Demand Forecasting Using Real-World Multi-Modal Data Integration (arXiv:2510.09048). *arXiv*. <https://doi.org/10.48550/arXiv.2510.09048>
10. Van Etten, T., Degeler, V., & Luo, D. (2024). Large-Scale Forecasting of Electric Vehicle Charging Demand Using Global Time Series Modeling: Proceedings of the 10th International Conference on Vehicle Technology and Intelligent Transport Systems, 40–51. <https://doi.org/10.5220/0012555400003702>

11. Wang, J., Kaushik, H. D., Jacob, R. A., & Zhang, J. (2025). Spatiotemporal planning of electric vehicle charging infrastructure: Demand estimation and grid-aware optimization under uncertainty. *iScience*, 28(9), 113368. <https://doi.org/10.1016/j.isci.2025.113368>
12. Wang, Z., Nie, Y., Sun, P., Nguyen, N. H., Mulvey, J. M., & Poor, H. V. (2023). ST-MLP: A cascaded spatio-temporal linear framework with channel-independence strategy for traffic forecasting [Preprint]. arXiv. <https://doi.org/10.48550/arXiv.2308.07496>